

**Multiple Choice:**

1. Given $P(A)=1/4$, $P(B)=1/3$ and $P(A \cup B)=1/2$, then what is the value of $P(B|A)$?
A. $1/2$ C. $1/4$
B. $1/6$ D. $1/3$
2. The probability that it will snow on Sunday is $3/5$. The probability that it will snow on both Sunday and Monday is $3/10$. What is the probability that it will snow on Monday, if it snowed on Sunday?
A. $1/2$ C. $1/4$
B. $1/6$ D. $1/3$
3. In how many different ways can the director of a research laboratory choose two chemical engineers from among five and two industrial engineers from among four applicants?
A. 30 C. 120
B. 60 D. 240
4. If a die is rolled 5 times, what is the probability of obtaining five 6s in a row?
A. 0.00129 C. 0.000129
B. 0.129 D. 0.00129
5. A box contains 5 defective and 195 non-defective cell phones. A quality control engineer selects two cell phones at random *with replacement*. What is the probability that both are defective?
A. 0.000625 C. 0.000550
B. 0.000450 D. 0.000725
6. Suppose a lot consisting of 100 items contains 5 defectives. If one item is randomly selected, what is the probability that it is defective?
A. 2.5 % C. 1.25 %
B. 0.5 % D. 5 %
7. A box contains 5 defective and 195 non-defective cell phones. A quality control engineer selects two cell phones at random with replacement. What is the probability that neither is defective?
A. 0.950625 C. 0.967200
B. 0.974932 D. 0.944800
8. Suppose 1% of parts have type 1 defect and 2% have type 2 defect and 0.5% have both types of defects. A part is known to have type 2 defect. What is the probability that it also has a type 1 defect?
A. 50 % C. 33 %
B. 67 % D. 25 %
9. Suppose a poker hand is dealt to an engineer at a casino and the engineer would like to know how many poker hands are possible when 5 cards are dealt from 52. How many different poker hands that the engineer could be dealt?
A. 2,986,960 C. 2,689,880
B. 2,790,450 D. 2,854,200
10. A bag contains 3 white and 5 black balls. If two balls are drawn in succession without replacement, what is the probability that both balls are black?
A. $5/16$ C. $5/28$
B. $5/32$ D. $5/14$
11. In a throw of two dice, the probability of obtaining a total of 10 or 12 is
A. $1/6$ C. $1/9$
B. $1/12$ D. $1/18$
12. Four different colored flags can be hung in a row to make coded signal. How many signals can be made if a signal consists of the display of one or more flags?
A. 64 C. 66
B. 68 D. 62
13. Find the probability of getting a prime number thrice by tossing a die 5 times.
A. 0.3125 D. 0.3421
14. In a certain college, 25% of the students failed in Mathematics, 15% failed in Chemistry and 10% failed both Mathematics and Chemistry. A student is selected at random. What is the probability that he failed neither Mathematics nor Chemistry?
A. 0.7 C. 0.4
B. 0.3 D. 0.6
15. An urn contains four black balls and six white balls. What is the probability of getting one black ball and one white ball in two consecutive draws from the urn?
A. 0.34 C. 0.53
B. 0.33 D. 0.12
16. If 15 people won prizes in the state lottery (assuming that there are no ties), how many ways can these 15 people win first, second, third, fourth and fifth prizes?
A. 340,360 C. 360,360
B. 360,306 D. 304,340
17. How many different signals, each consisting of eight flags hung one above the other, can be formed from a set of three indistinguishable red flags, two indistinguishable blue flags, two indistinguishable white flags and one black flag?
A. 1390 C. 1200
B. 1680 D. 1900
18. Which of the following are events that are dependent?
A. Tossing a coin and rolling a die
B. Rolling a die and then rolling another die.
C. Drawing a card from a deck and rolling a die
D. Sunbathing all day and getting a sunburn
19. Three screws are drawn at random from a lot of 100 screws, 10 of which are defective. Find the probability that the screws drawn will be non-



- defective in drawing without replacement.
- A. 0.6275 C. 0.4567
B. 0.7265 D. 0.5067
20. From six history books and eight economics books, in how many ways can a person select two history books and three economics book and arrange them on a shelf?
- A. 100,800 C. 120,000
B. 130,400 D. 108,000
21. How many three letter words (without meaning) are possible when repetition of letters is not allowed? What is the probability that those words will not start with a vowel?
- A. 0.2461 C. 0.50
B. 0.8077 D. 0.345
22. The probability that A wins a certain game is $\frac{2}{3}$. If A plays 5 games, what is the probability that A will win exactly 3 games?
- A. $\frac{21}{121}$ C. $\frac{80}{243}$
B. $\frac{3}{130}$ D. $\frac{56}{111}$
23. A coin is tossed 3 times and 2 heads and 1 tail fall. What is the probability that the first toss was head?
- A. $\frac{2}{3}$ C. $\frac{1}{4}$
B. $\frac{3}{4}$ D. $\frac{1}{2}$
24. Find the number of ways 2 balls, 4 dolls and 6 toy guns can be given to 12 children if each child gets a toy.
- A. 13,860 C. 14,200
B. 12,657 D. 11,890
25. Which of the following are events that are mutually exclusive?
- A. Drawing a card from a deck and getting a queen and a heart.
B. Rolling a die and getting an even number and a 2.
C. Rolling two dice and getting doubles or getting a sum of 6.
D. Tossing two coins and getting two heads or two tails.
26. The population of Nicosia (Cyprus) is 75 % Greek and 25% Turkish. 20% of the Greeks and 10% of the Turks speak English. If the person the visitor met speaks English, what is the probability that the person is a Greek?
- A. 0.857 C. 0.668
B. 0.725 D. 0.586
27. In a certain communication system, a word consists of a sequence of 3 elements, where each element may take the value of 1 or 2. If each word is equally like to happen, the probability that a word occurring with exactly two elements of value 1 is ____.
- A. 0.375 C. 0.250
B. 0.325 D. 0.425
28. Which of the following are events that are not mutually exclusive?
- A. Tossing a coin and getting a tail and rolling a die and getting an even number.
B. Tossing a coin and getting a head or a tail.
C. Rolling a die and getting a 6 or a 3.
D. Drawing a card from a deck and getting a club and an ace
29. It is given that the events A and B are such that $P(A) = \frac{1}{4}$, $P(A|B) = \frac{1}{2}$ and $P(B|A) = \frac{2}{3}$. Find $P(B)$.
- A. $\frac{2}{3}$ C. $\frac{1}{6}$
B. $\frac{1}{2}$ D. $\frac{1}{3}$
30. Suppose A and B are events with $P(A) = 0.6$, $P(B) = 0.3$ and $P(A \cap B) = 0.2$. Find the probability that neither A nor B occurs.
- A. 0.7 C. 0.4
B. 0.3 D. 0.6
31. How many distinguishable permutations are there of the letters in MISSISSIPPI?
- A. 35,640 C. 36,540
B. 34,650 D. 30,654
32. In how many ways can 6 people be seated around a circular table?
- A. 60 C. 240
B. 120 D. 360
33. If seven coins are tossed simultaneously, find the probability that there will be at least six tails.
- A. $\frac{1}{3}$ C. $\frac{1}{15}$
B. $\frac{1}{16}$ D. $\frac{1}{4}$
34. If a fair coin is flipped 20 times, what is the probability of the coin landing on heads exactly 10 times?
- A. 0.123 C. 0.154
B. 0.176 D. 0.198
35. Sixty percent of the people who buy sports cars are men. If 10 sports car owners are randomly selected, find the probability that exactly 7 are men.
- A. 0.314 C. 0.450
B. 0.200 D. 0.215
36. In a shooting match, eight clay targets are arranged in two hanging columns of three targets each and one column of two targets. A marksman is to break all the targets according to the certain rules. First the marksman chooses a column from which a target is to be broken. Then the marksman must then break the lowest remaining target in the chosen column. In how many different orders can the eight targets be broken, given that the rules are followed?
- A. 240 C. 560
B. 380 D. 440
37. What is the probability that at least two students of a class of size 23 have the same birthday?
- A. 0.316 C. 0.612



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Math 02-01 to Math 02-13
Probability

- B. 0.442 D. 0.507
38. From a bag containing 4 black balls and 5 white balls, two balls are drawn one at a time. Find the probability that one ball is white and one ball is black. Assume that the first ball is returned before the second ball is drawn.
- A. 16/81 C. 25/81
B. 20/18 D. 40/81
39. In an Engineering Board Examinations, the probability that an Excel examinee will pass each subject is 0.8. What is the probability that an Excel examinee will pass at least two subjects out of the three board subjects?
- A. 70.9 % C. 80.9 %
B. 85.9 % D. 89.6 %
40. A fair coin is tossed three times. What is the probability of getting either 3 heads or 3 tails?
- A. 1/8 C. 3/8
B. 1/4 D. 1/2
41. How many committees can be formed by choosing 4 men from an organization of a membership of 15 men?
- A. 1390 C. 1240
B. 1435 D. 1365
42. What is the number of permutations of the letters in the word BANANA?
- A. 36 C. 60
B. 52 D. 42
43. Determine the probability of drawing either a king or a diamond in a single draw from a pack of 52 playing cards.
- A. 2/13 C. 3/13
B. 4/13 D. 1/13
44. An urn contains 4 black balls and 6 white balls. What is the probability of getting 1 black and 1 white ball in two consecutive draws from the urn?
- A. 0.24 C. 0.27
B. 0.53 D. 0.04
45. A plant has three suppliers. S1 supplies 30% of the parts to the plant, S2 supplies 50% of the parts, and S3 supplies the remaining 20%. One percent of the parts supplied by S1 are defective, two percent of the parts supplied by S2 are defective, and three percent of the parts supplied by S3 are defective. What percentage of parts at the plant are defective?
- A. 1.9 % C. 2.0 %
B. 2.1 % D. 2.2 %
46. Suppose only 12% of men in ancient Greece were honest. What is the probability that the first honest man Diogenes encounters will be the third man he meets?
- A. 0.068 C. 0.074
B. 0.045 D. 0.093
47. Super Mario cards were in one-third of cereal boxes advertising enclosed Magic Motion cards. If six boxes of cereal are purchased, what is the probability of exactly two Super Mario cards?
- A. 0.438 C. 0.329
B. 0.526 D. 0.265
48. The probability that Excel's bus is on time is $\frac{2}{3}$, and the probability that Mr. Alarde is driving the bus is $\frac{4}{5}$. What is the probability that on any given day Excel's bus is on time and Mr. Alarde is the driver?
- A. $\frac{7}{15}$ C. $\frac{9}{15}$
B. $\frac{8}{15}$ D. $\frac{6}{15}$
49. Selena and Tracey play on a softball team. Selena has 8 hits out of 20 times at bat, and Tracery has 6 hits out of 16 times at bat. Based on their past performance, what is the probability that both girls will get a hit next time at bat?
50. A hand soap manufacturer introduced a new liquid, lotion-enriched antibacterial soap and conducted an extensive consumer survey to help judge the success of the new product. The survey showed 40% of the consumers had seen an advertisement for the new soap, 20% has tried the new soap and 15% had both seen an advertisement and tried the new soap. If a randomly selected consumer has seen an advertisement for the new soap, what is the probability that this consumer has tried the new soap?
- A. 72% C. 25%
B. 40% D. 37.5%
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